Product Information
Acronal® 7054

Polymer dispersion for high quality, all purpose architectural coatings

Acronal® 7054 is an anionic polymer dispersion with a medium viscosity. It consists of small, finely divided particles. Its compatibility with pigments and extenders is excellent, and it has very high binding power for pigments. Unpigmented films of Acronal® 7054 are tack free at room temperature. Films formed by Acronal® 7054 are clear, flexible and glossy, they have extremely high water resistance, and they are very resistant to saponification.

Chemical Nature:
Polymer dispersion of acrylic ester and styrene

Benefits

- Ability to formulate a wide range of architectural coating applications for interior and exterior
- Long lasting protective finish
- Excellent paint properties with lower binder content
- Outstanding processing properties

Features

- Versatile polymer dispersion easy to formulate
- Best in class weatherability
- High pigment binding power
- Excellent mechanical and electrolyte stability

Properties

Product specification*	Solids content	%	45 ± 1
	pH value	pH	8.0 – 10.5
	Viscosity at 23 °C, LV DV 3/100	mPa s	50 – 900
Other properties of dispersions	Minimum film-forming temperature (ISO 2115)	°C	18
	Density (ISO 2811-1)	g/cm^3	approx. 1.04
	Resistance to frost	°C	≤ 0
	Type of dispersion		anionic

*The aforementioned data shall constitute the agreed contractual quality of the product at the time of passing of risk. The data are controlled at regular intervals as part of our quality assurance program. Neither these data nor the properties of product specimens shall imply any legally binding guarantee of certain properties or of fitness for a specific purpose. No liability of ours can be derived therefrom.

Note
The information submitted in this publication is based on our current knowledge and experience. In view of the many factors that may affect processing and application, these data do not relieve processors of the responsibility of carrying out their own tests and experiments; neither do they imply any legally binding assurance of certain properties or of suitability for a specific purpose. It is the responsibility of those to whom we supply our products to ensure that any proprietary rights and existing laws and legislation are observed.

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Applications

Areas of application

All purpose binder, mainly used for the following applications:

- Exterior paints
- Interior paints, plastic paints for interior application
- Textured finishes
- Primers for mineral substrates
- Modifying silicate paints and finishes

Processing

It is advisable to disperse the pigments and extenders with wetting and dispersing agents such as Dispex® AA 4030 and water-soluble polyphosphates in an alkaline medium in advance before the polymer dispersion is added. It is only when products with very high viscosity are being mixed in low-speed mixers that Acronal® 7054 should be added together with the wetting and dispersing agents.

Acronal® 7054 has high pigment binding power, and very good compatibility with pigments and fillers.

Various thickeners can be added to emulsion paints in order to adjust their viscosity and workability. Cellulose ethers, polyacrylates, diurethane thickener (such as Rheovis® AS 1125) and bentonite can be used. The choice of thickener depends on whether the coating is expected to be free-flowing or more thixotropic.

Solvents need to be added in order to enable the polymer to form a uniform film at temperatures below ambient temperature. It is usually sufficient to add these solvents at a level of 5 - 8%, expressed as a proportion of the total solid content of dispersion. Short-chain alcohols and glycols improve the freeze-thaw resistance of paints, but they cannot be used to lower the film-forming temperature. If possible, solvents should not be added direct to the polymer dispersion, they should be mixed with the pigment paste and then added.

Like all finely divided polymer dispersions, Acronal® 7054 has a tendency to foam. It is therefore necessary to add a commercial defoamer at the level of 0.3 – 0.6 %. Trials should be carried out to test the effectiveness of the defoamer.

Although Acronal® 7054 itself is resistant to microorganisms in the form in which it is supplied, preservatives need to be added to products formulated with Acronal® 7054 to protect them from attack by micro organisms over long periods in storage. Trials should always be carried out to test the compatibility and efficacy of the preservatives.

Customers have to carry out their own trials when developing and processing products based on Acronal® 7054. The compatibility of Acronal® 7054 with other ingredients of formulations, its effect on mixing processes and its adhesion on different substrates etc., are affected by a variety of factors which are too numerous for us to take into account in our own trials. This includes testing its stability by storing it at ca. 50 °C to confirm that its viscosity remains stable.

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Safety

General
The usual precautions for handling chemicals must be observed. These include the measures set out in the guidelines of the organizations responsible for safety at work, in particular, good ventilation and fume extraction at the workplace, care of the skin and the wearing of eye protection.

Safety Data Sheet
When using this product, the information and advice given in our **Safety Data Sheet** should be observed. Due attention should also be given to the **precautions** necessary for handling chemicals.

Labeling
According to all the data at our disposal, Acronal® 7054 does not need to be labelled as a dangerous substance or preparation as defined in the relevant local directives according to their current status.

Storage

Acronal® 7054 must not be allowed to come into contact during storage with metals or alloys that are susceptible to corrosion. It is important to ensure that containers are kept tightly sealed, and the headspace of bulk storage tanks must be kept saturated with water vapor. This product must not be exposed to high temperatures, and it must be protected from frost.

Acronal® 7054 has a shelf life of nine months at 10 – 30 °C, provided due attention is paid to the hygiene of tanks and storage facilities.

We would recommend treating this product with a biocide in order to prevent problems with microorganisms from occurring during storage and processing.

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